

Understanding the Impact of Economic, Demographic, Energy Factors & Plastics on Global Warming

Springboard Capstone II Project Report
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BLUE PLANET II



This continues to be a problem because.....

01

Global CO₂ Emissions are up 60% in past 3 decades)

02

Global Plastic Production up ~280% in past 3 decades

03

Global Energy Demand continues to increase as well

04

Energy derived from Fossil Fuels leads to Global Warming

Problem Statement



**NATIONAL
ECONOMIC
ACTIVITY**



**ENERGY
USAGE**



**DEMOGRAPHIC
INDICATORS**



**PLASTIC INTENSIVE
GLOBAL SUPPLY
CHAINS**



**CARBON DIOXIDE
EMISSION LEVELS**

Key Indicator of
Global Warming



**PREDICT & PROVIDE
ACTIONABLE
INSIGHTS**



1990s



2000s



2010s



2020s



**PREDICT FUTURE
CO₂ TRENDS**



**QUANTIFY IMPACT
OF KEY DRIVERS**



**INFORM POLICIES &
SUPPLY CHAIN
STRATEGIES**



**SUPPORT SUSTAINABLE
DECISION MAKING**

Analysis Objectives

- Identify Key Drivers of CO2 Emissions
- Compare Total CO2 Emissions Vs. CO2 Per Capita
- Contextualize Findings within Rising Plastics Production
- Develop Predictive Machine Learning Models

Datasets Used

The data was acquired from <https://ourworldindata.org/>

1. Carbon Emissions Data (Country Year Level)

<https://raw.githubusercontent.com/owid/co2-data/master/owid-co2-data.csv>

Citation : Our World in Data. *CO₂ and Greenhouse Gas Emissions* [dataset]. Data adapted from the Global Carbon Project. Retrieved from <https://github.com/owid/co2-data> (accessed 2026).

2. Global Plastics Production Data (Global Year Level)

<https://ourworldindata.org/grapher/global-plastics-production>

Citation : Geyer, R., Jambeck, J., and Law, K. (2017) – with processing by Our World in Data. *Global plastics production* [dataset]. Retrieved from <https://ourworldindata.org/grapher/global-plastics-production> (accessed 2026).

Data Wrangling

Key Steps

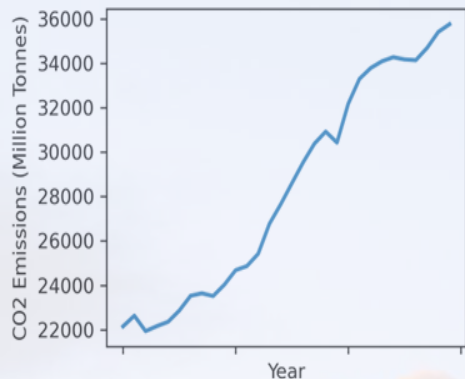
- Reduced CO2 Dataset from 79 to 10 relevant columns
- Removed non-country entries
- Filtered both Plastics and CO2 Datasets for 1990-2019 Timeframe
- Handled missing values
- Created merged Global CO2 + Plastics Dataset
- Standardized Units

Final Datasets

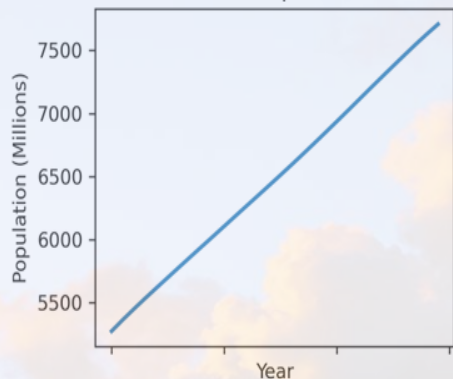
- CO2 Master Country Year Level Dataset for Modeling
- Global CO2 + Plastics Dataset for Trends Analysis

Global Trends

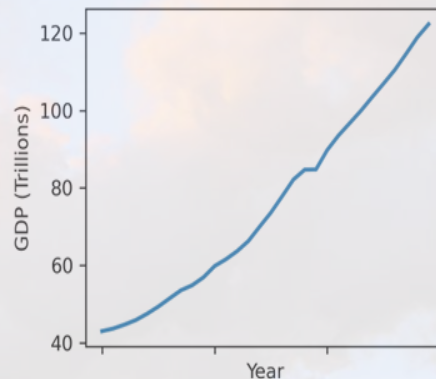
Global CO2 Emissions



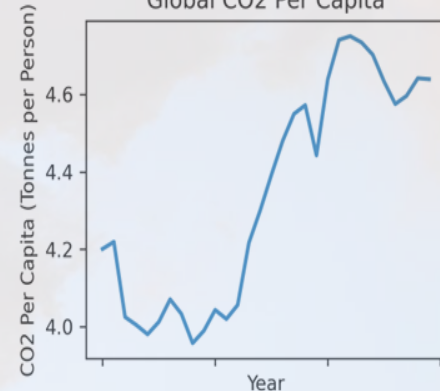
Global Population



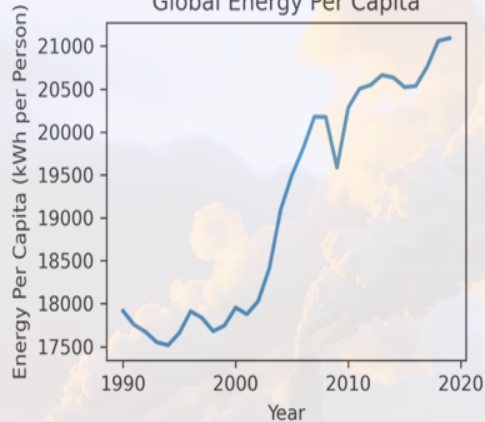
Global GDP



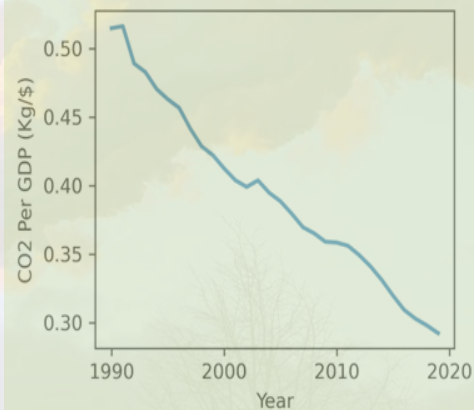
Global CO2 Per Capita



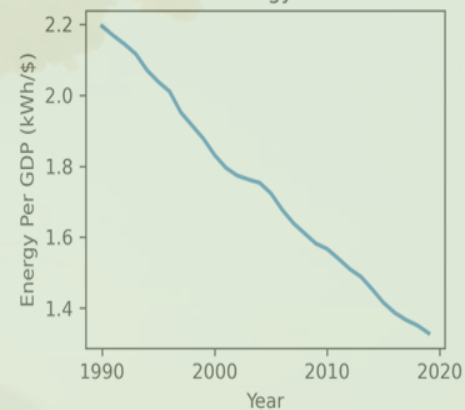
Global Energy Per Capita



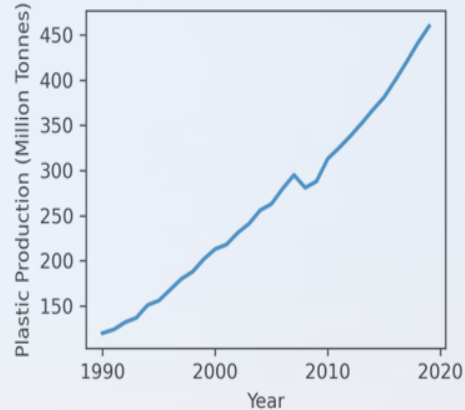
Global CO2 Per GDP



Global Energy Per GDP



Global Plastic Production



Exploratory Data Analysis

Key Insights

- Energy per capita is strongly correlated with CO2 Emissions
- Higher GDP → Higher CO2 Emissions
- Higher Population Size → Higher Total CO2 Emissions
- Developed countries show signs of decoupling

Important Observation

- Emissions stabilized for some developed countries despite economic growth

Feature Engineering & Preprocessing

Key Steps

- Log Transformation of Skewed Features
- PCA Transformation of Features for Multicollinearity
- Standardization of Features
- Time aware Train/Test Splits for 2 Modeling Approaches
 - Total CO2 Emissions as Target
 - CO2 per Capita as Target

Modeling Strategy

Modeling Approaches

Approach	Target Variable	Features
1 st Strategy	log_co2	log_energy_per_capita, log_gdp_per_capita, log_population
2 nd Strategy	log_co2_per_capita	log_energy_per_capita, log_gdp_per_capita

Models Used : Ordinary Least Squares with country fixed effects (Scaled & Unscaled), Scaled Ridge Regression , Random Forest Model

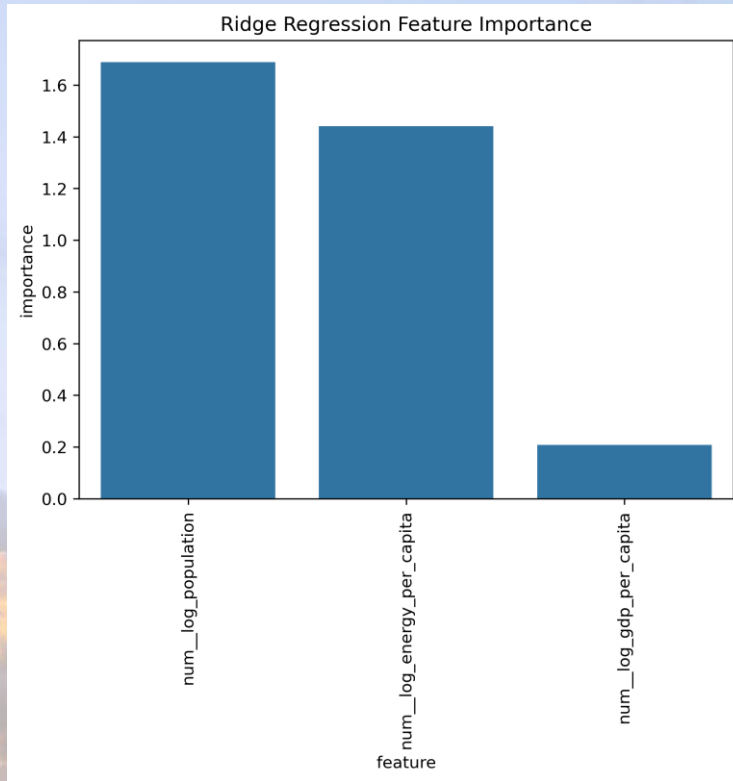
Evaluation Metrics : R^2 , RMSE, MAE

Model Performance Summary & Selection

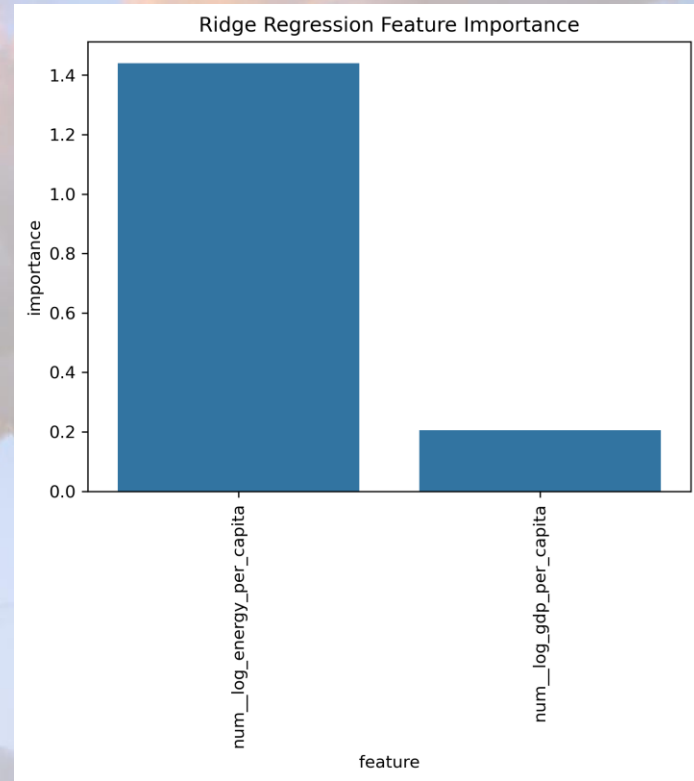
Target	Models	R2	RMSE	MAE	Final Model	Comments
Total CO2 Emissions	Ordinary Least Squares Fixed Effects (Unscaled)	0.985	0.263	0.201	No	Interpretable, Strong Performance due to Population Signal Dominance
Total CO2 Emissions	Ordinary Least Squares Fixed Effects (Scaled)	0.985	0.263	0.201	No	Interpretable, Strong Performance due to Population Signal Dominance
Total CO2 Emissions	Ridge Regression (Scaled)	0.988	0.231	0.167	No	Strongest Performance, Interpretability & Stability
Total CO2 Emissions	Random Forest (Scaled)	0.98	0.301	0.205	No	Lower Performance but captures Non Linearity
CO2 Per Capita	Ordinary Least Squares Fixed Effects (Unscaled)	0.969	0.263	0.201	No	Lower Performance with Population Signal removed
CO2 Per Capita	Ordinary Least Squares Fixed Effects (Scaled)	0.969	0.263	0.201	No	Lower Performance with Population Signal removed
CO2 Per Capita	Ridge Regression (Scaled)	0.976	0.231	0.167	Yes	Strongest Performance, Interpretability & Stability - Final Selection as it is not driven by population
CO2 Per Capita	Random Forest (Scaled)	0.973	0.247	0.182	No	Good Performance and Confirms Feature Importance from Ridge Model above

Feature Importance

With Population Signal for Total CO2 Emissions

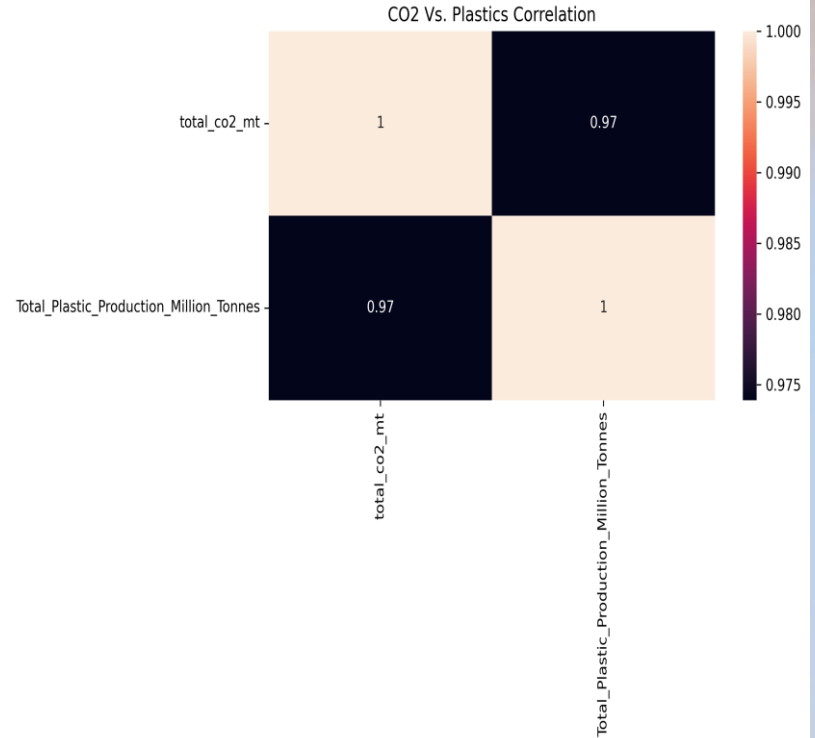
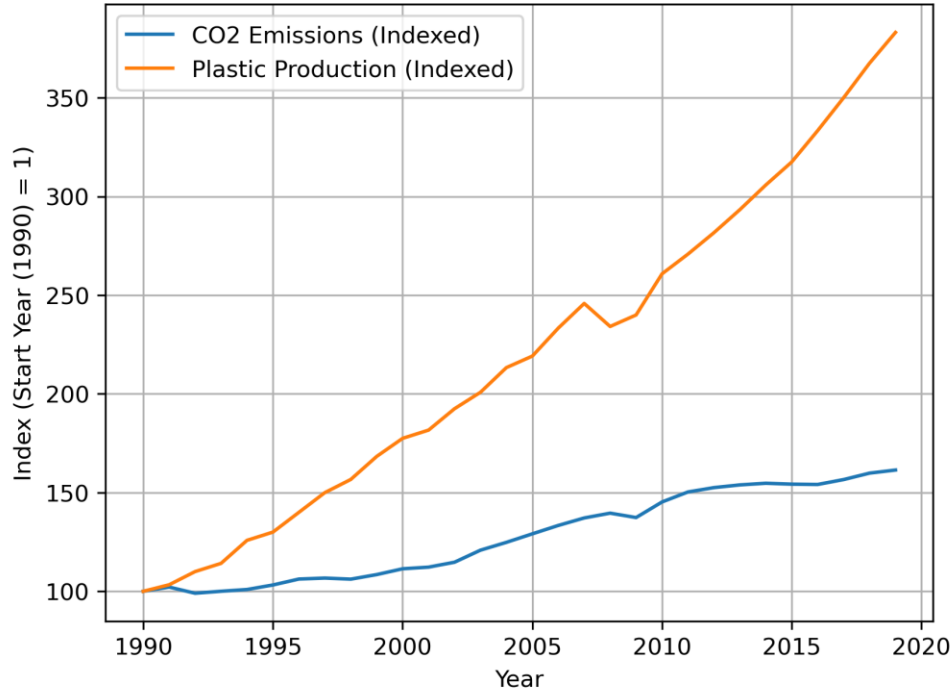


Without Population Signal for CO2 per Capita



Plastics Connection

Global CO2 Emissions vs. Plastic Production Trend Over Past Decades



Key Findings

- Population Size drives Total CO2 Emissions
- Energy Demand drives CO2 Emissions Intensity
- GDP drives CO2 Emissions indirectly via Energy Demand
- Decoupling of Emissions with Economic Growth exists but not always
- Plastic Intensive Supply Chains increase environmental pressure

Actionable Recommendations

- Track both CO2 per capita and CO2 Emissions for all major energy intensive industries & processes within a country
- Invest in energy efficient technologies
- Incentivize green/renewable energy sources
- Invest in recycling and improving Supply Chain sustainability
- Reduce dependence on plastics by investing in greener alternatives

Future Work

- Add country/year level plastics data
- Add renewable energy indicators
- Explore panel regression models
- Build forecasting dashboard

Conclusion

- Energy consumption behavior has the highest environmental impact
- Energy Efficiency gains offset by scale of growth, energy demand, population and rapidly increasing plastic supply chains

Final Insight

- Global Warming mitigation will not just require cleaner energy systems but also reducing material intensity for global supply chains

Thank You!

Questions?

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- GitHub Repository : <https://github.com/gawandesaillee-lab/learndsp/blob/main/Data%20Science%20Capstone%202%20Project/reports/Data%20Science%20Capstone%202%20Project%20Report.pdf>